

Introduction

SmallSats have transformed access to space and are widely used for Earth imaging, technology demonstrations, climate monitoring, communications, and The design of SmallSats is heavily shaped by strict mass, volume, and budget limitations, leading to highly standardized geometries and simplified structural configurations.

Rationale

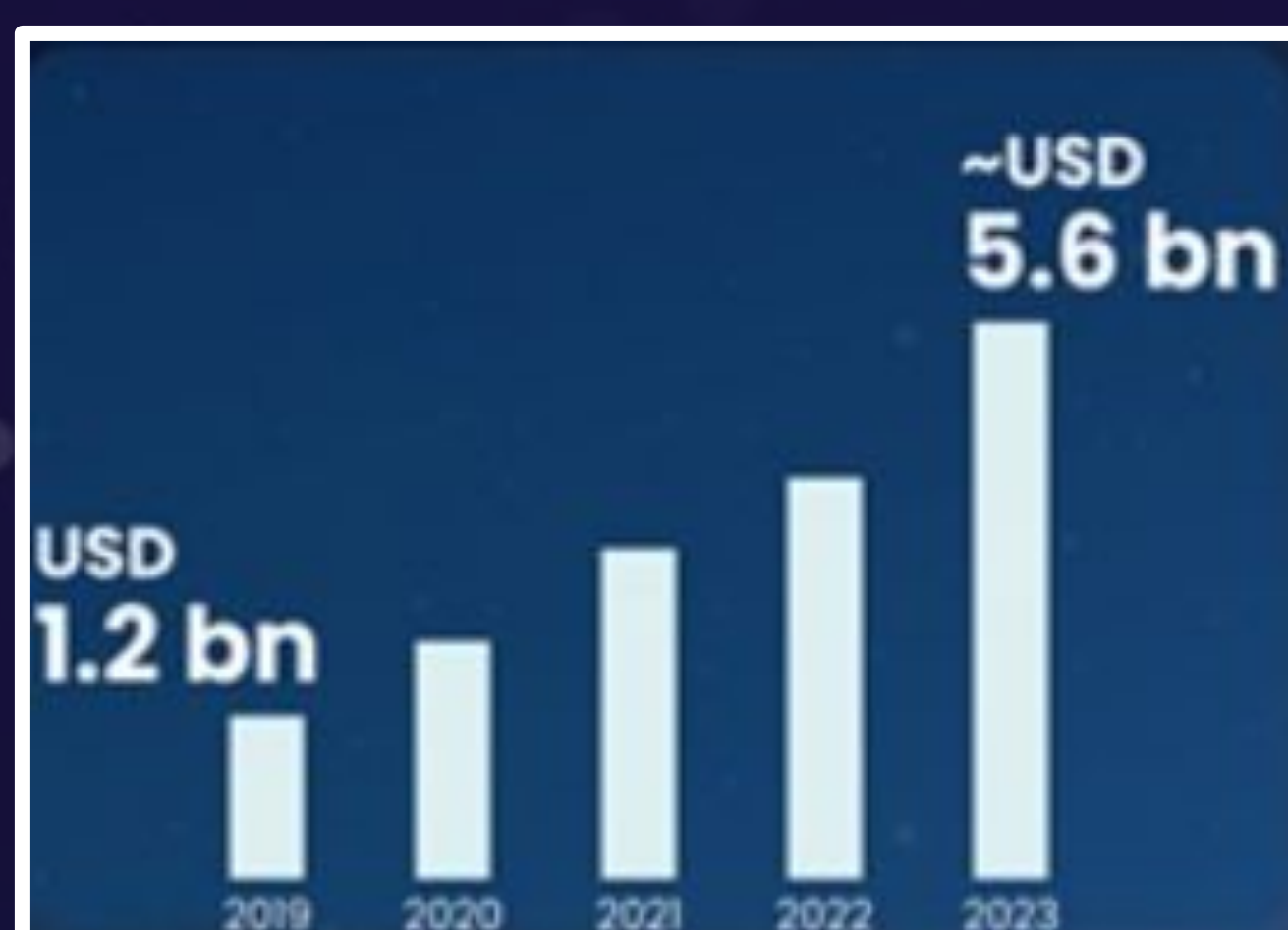


Fig 1. Market for SmallSats over Time (Stratview Research 2024)

Over 97% of small satellites burn up during reentry, causing the loss of billions of dollars in hardware and data each year and increasing space debris. This limits the recovery of experiments, reuse of expensive sensors, and execution of repeated or time-sensitive missions.

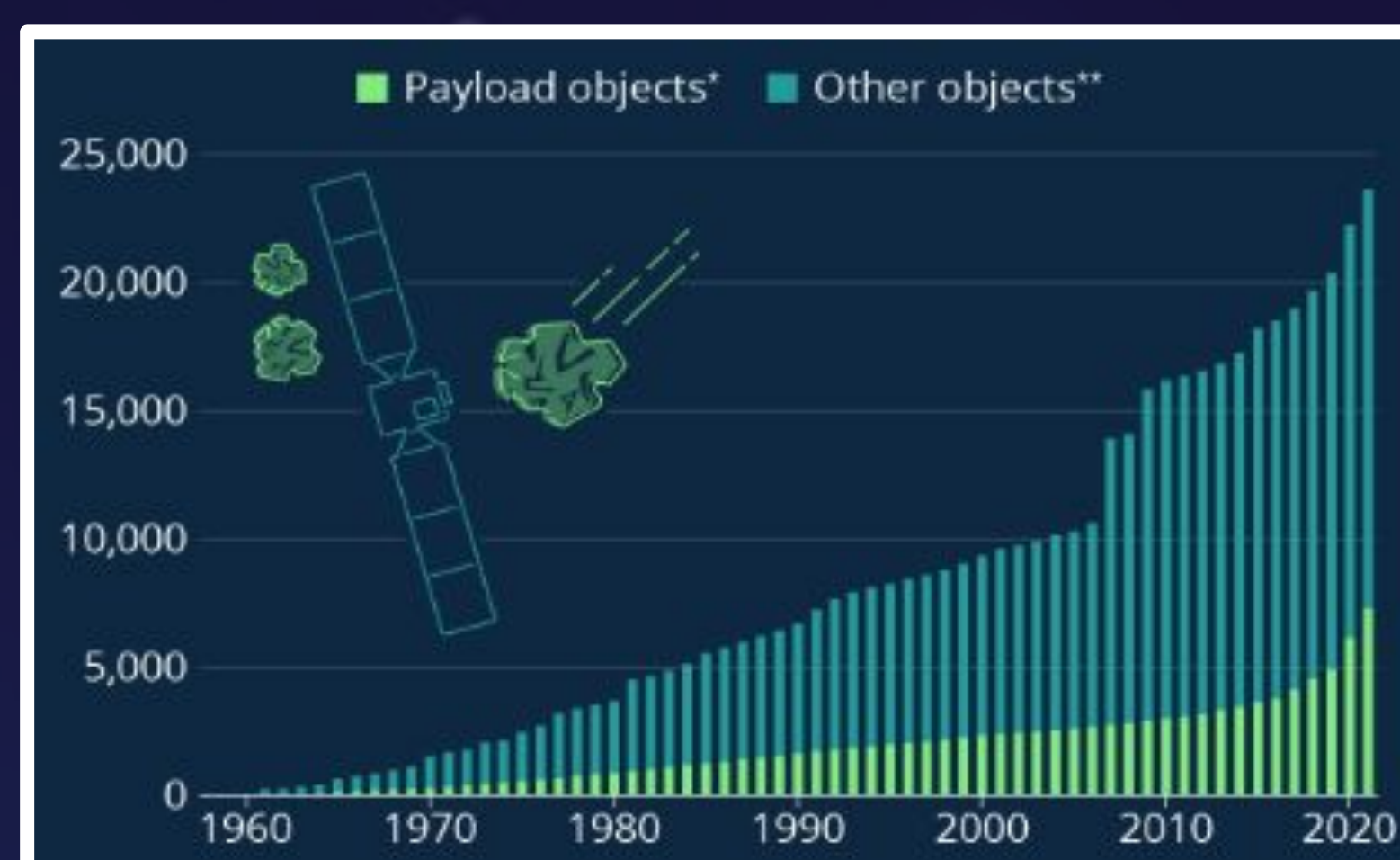


Fig 2. Space Debris over Time (General Catalogue of Artificial Space Objects 2021)

A recoverable reentry system would improve sustainability by enabling payload reuse, lowering operational costs, and reducing environmental impacts. By developing a shape-optimized, thermally survivable, stable nose cone, companies and people will have an affordable option to recover and reuse their smallsats

Optimization of Nose Cones for Atmospheric Reentry of SmallSATS

Methods

1



Fig 3. Five Nose Cone Geometries that were Tested

5 shape families were modeled and tested: sphere cone, blunted biconic, blunted ogive, power-law, and hyperbolic. Each were ran in CFD simulations to measure static stability (using center of pressure) and max heating (using wall heat flux) to find the optimal family

2

The optimal shape was modeled and tested at 5 different heights from 0.1m to 0.5m. Each were ran in the same simulations as phase 1 to measure static stability and max heating to find the half angle of the shape family

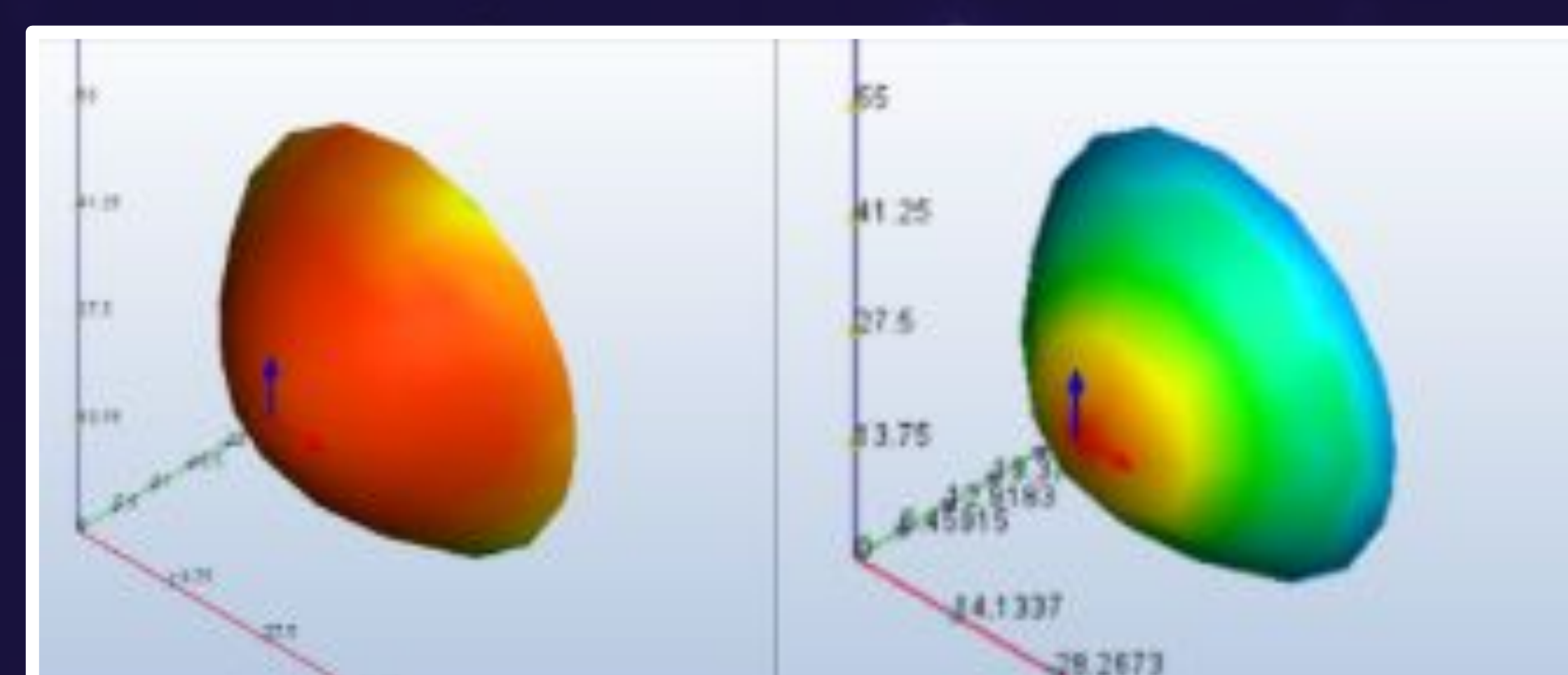


Fig 4. Examples of Wall Heat Flux (Left) and Static Stability (Right) Simulations

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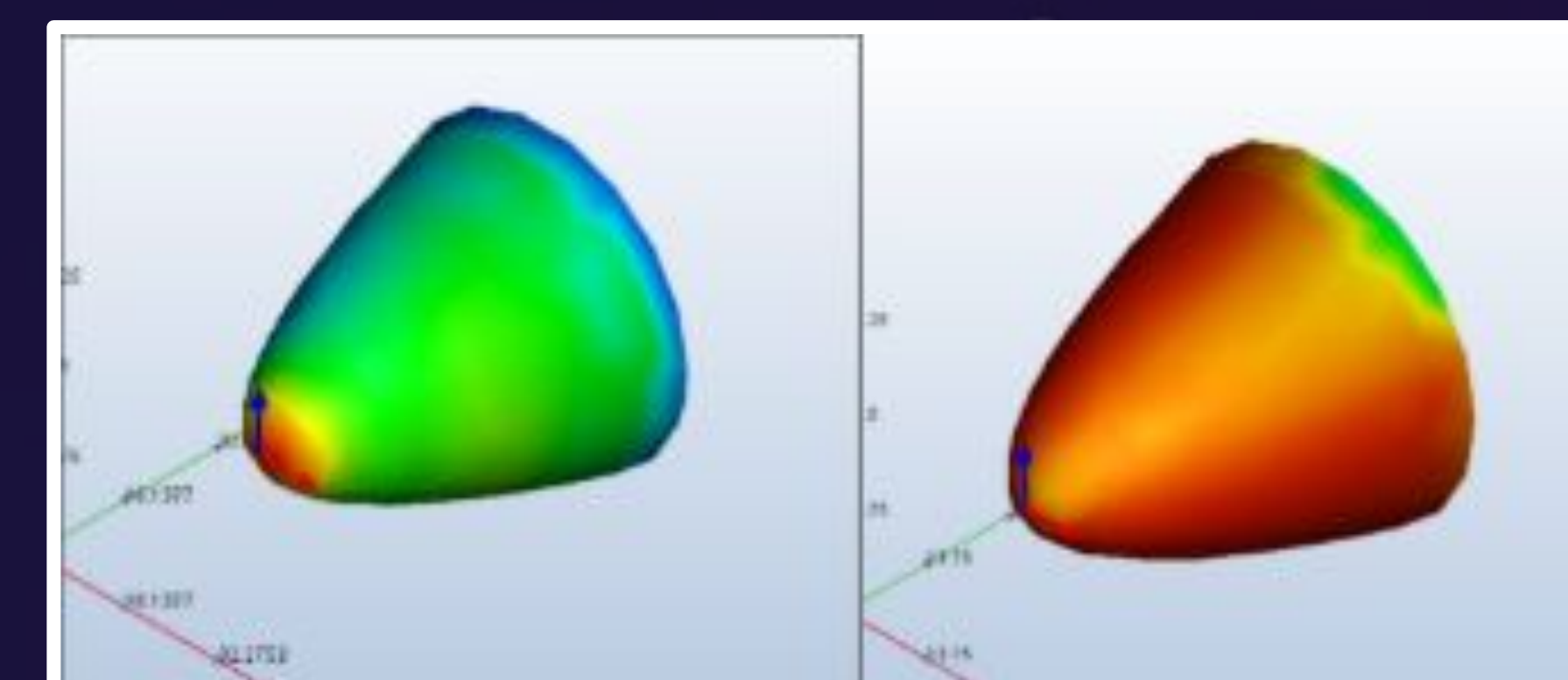


Fig 5. Examples of Max Heating (Right) and Dynamic Stability (Left) Simulations

The final shape was run through CFD simulations at an angle of attack sweep to determine dynamic stability and max speed to better determine surface heating. Orbital decay simulations were then run to determine descent path control.

Results

Cone	Static Margin (cal)	Wall Heat Flux (W/m^2)
Sphere cone	0.272	2359.029725
biconic	0.046	482.8718846
Blunted Ogive	0.28	8028.963415
Power-law	0.32852	257.5397407
Hyperbolic	0.202	5826.866507

Fig 6. Static Stability and Heating for Shape Optimization

Cone Height (m)	Static Margin (cal)	Wall Heat Flux (W/m^2)
0.5	0.32852	0.65704
0.4	0.25052	0.6263
0.3	0.20888	0.696266667
0.2	0.134318	0.67159
0.1	0.06274	0.6274

Fig 7. Static Stability and Heating for Height Optimization

Average Max Heat Flux (W/m^2)	122513.8853
Estimated Surface Temperature (°C)	1007.62236

Fig 8. Peak Surface Temperature Calculation for Material Determination

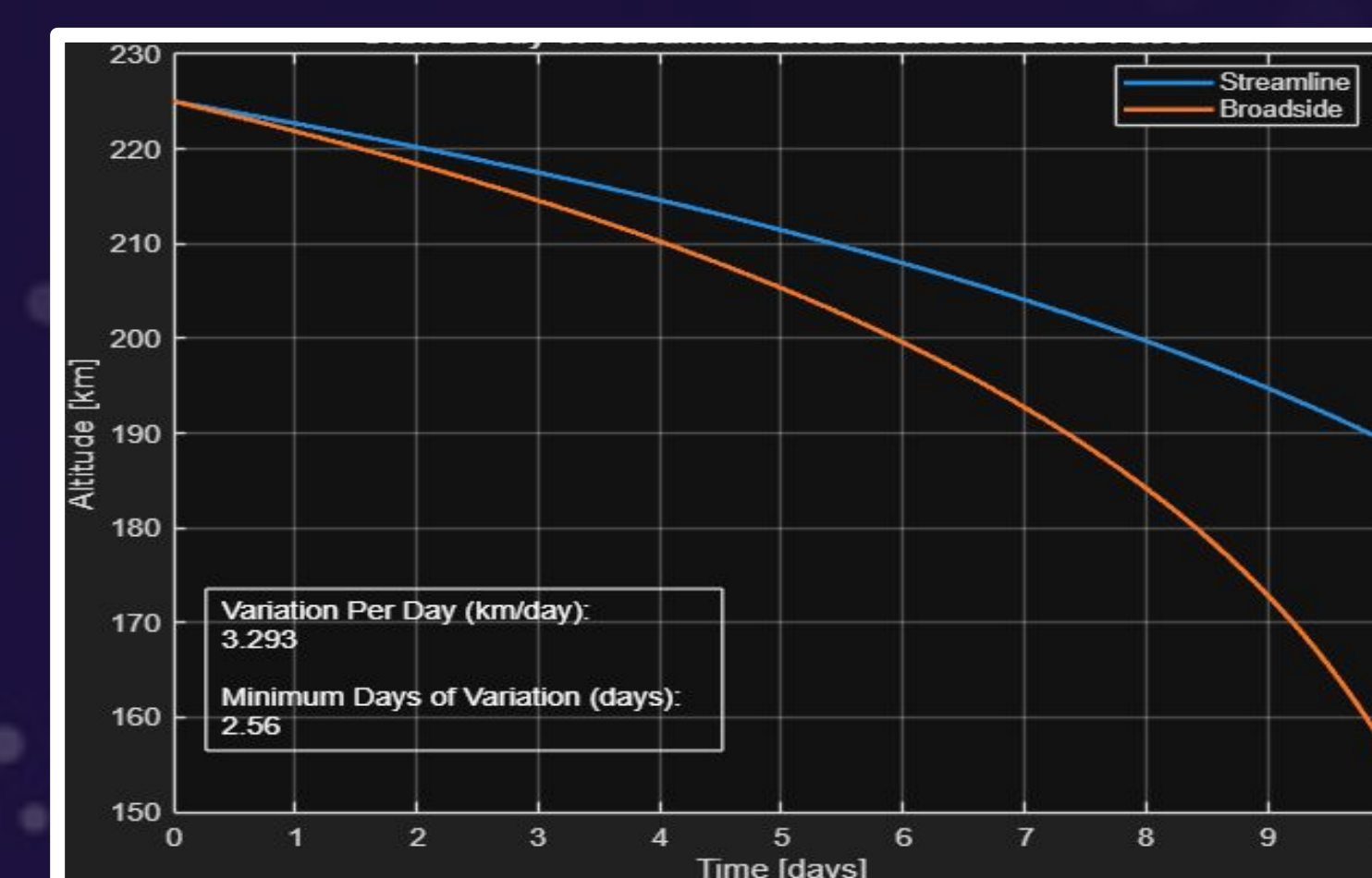


Fig 9. Orbital Decay Based on Satellite Orientation to Determine Splashdown Controllability



Fig 10. Pitching Moment Coefficient over AoA to Determine Satellite Dynamic Stability

Conclusion

The data in Figures 6 and 7 show that a 0.5 m power-law cone is the optimal configuration, minimizing wall heat flux while maximizing static margin. Figure 10 confirms dynamic stability, as the slightly negative pitching moment vs. AoA slope indicates restoring torques. Figure 8 shows a maximum surface temperature of about 1000°C, which is within the limits of the chosen Nextel 440 and alumina-silica ceramic fiber blanket shell, keeping material costs around \$1500. Finally, Figure 9 demonstrates that onboard control systems, such as reaction wheels, enable multi-day variation in touchdown timing to allow for safe landing. Overall, the design is viable due to its low cost, strong stability, low thermal loads, and landing control, making it a practical, reusable smallsat reentry solution.

Extensions

Future work will include a more detailed cost analysis, including manufacturing processes, material sourcing, scalability, and integration of subsystems to better estimate total system cost. In addition, physical prototyping and high-temperature materials testing will be conducted to ensure thermal performance, structural integrity, and real-world behavior under reentry-like conditions.

References

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